pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

ROC Curves and Choosing a Threshold

WEEK 2 · DAY 3

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Where day 7 left off

Day 7: threshold 0.5

324 of 1,327

defaults caught, 6,000 validation rows

Recall 0.2442, precision 0.7248. Nothing about 0.5 says it is the place to stop.

Real data

UCI card file: accounts in Taiwan, 2005. Not US credit today.

People's attributes

SEX, EDUCATION, MARRIAGE, AGE: never features.

60 / 20 / 20

Training fits, validation chooses, test reports once (day 10).

Names

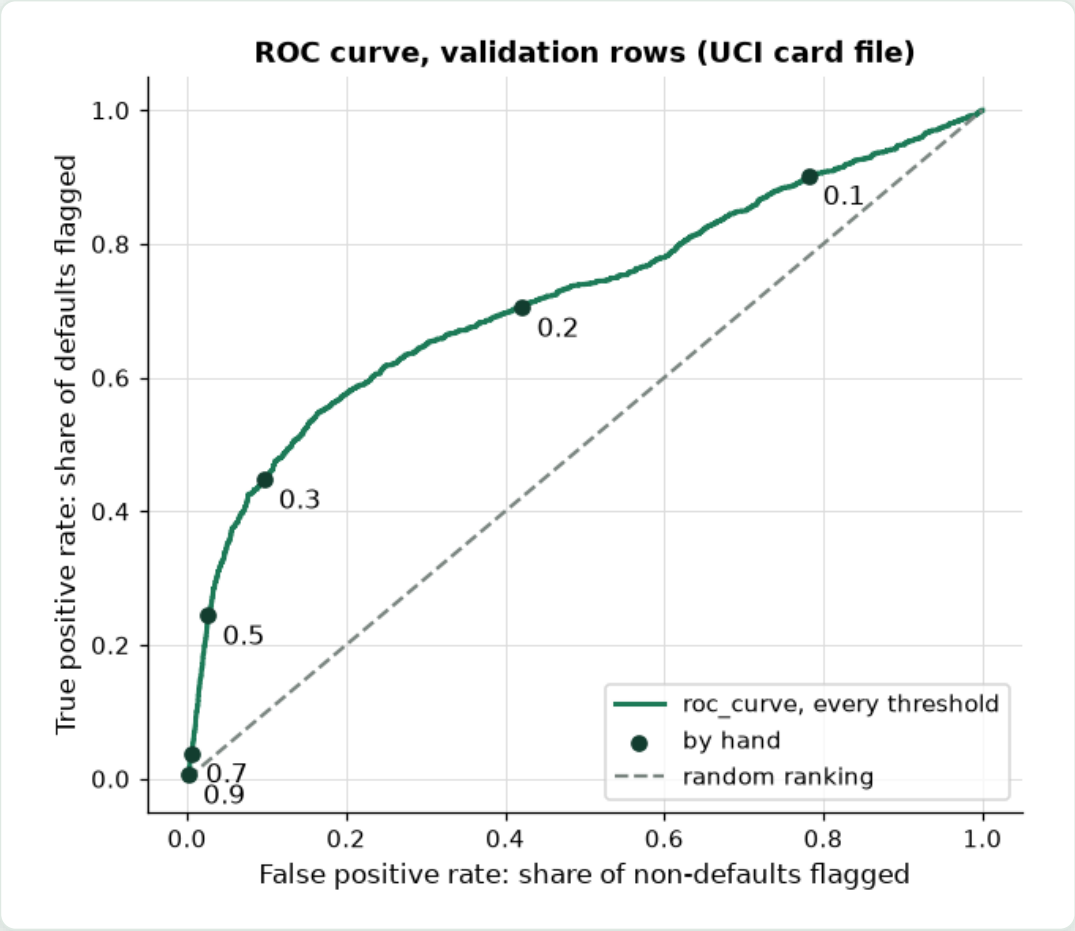
Model probability of default in this file; flagged at threshold t.

Every threshold and target today is a teaching convention, not what a lender does or should do.

How the ROC curve works

Every threshold	Each threshold turns the probabilities into a different set of flags
Two rates	True positive rate: the share of defaults flagged. False positive rate: the share of non-defaults flagged
The plot	One point per threshold, from (0, 0) where nothing is flagged to (1, 1) where everything is
The reading	A curve bowing toward the top left catches many defaults for few false flags
You control	Which rows to score

Every threshold at once: the ROC curve



Rate	Formula	Out of
True positive (recall)	TP / (TP + FN)	1,327 defaults
False positive	FP / (FP + TN)	4,673 others

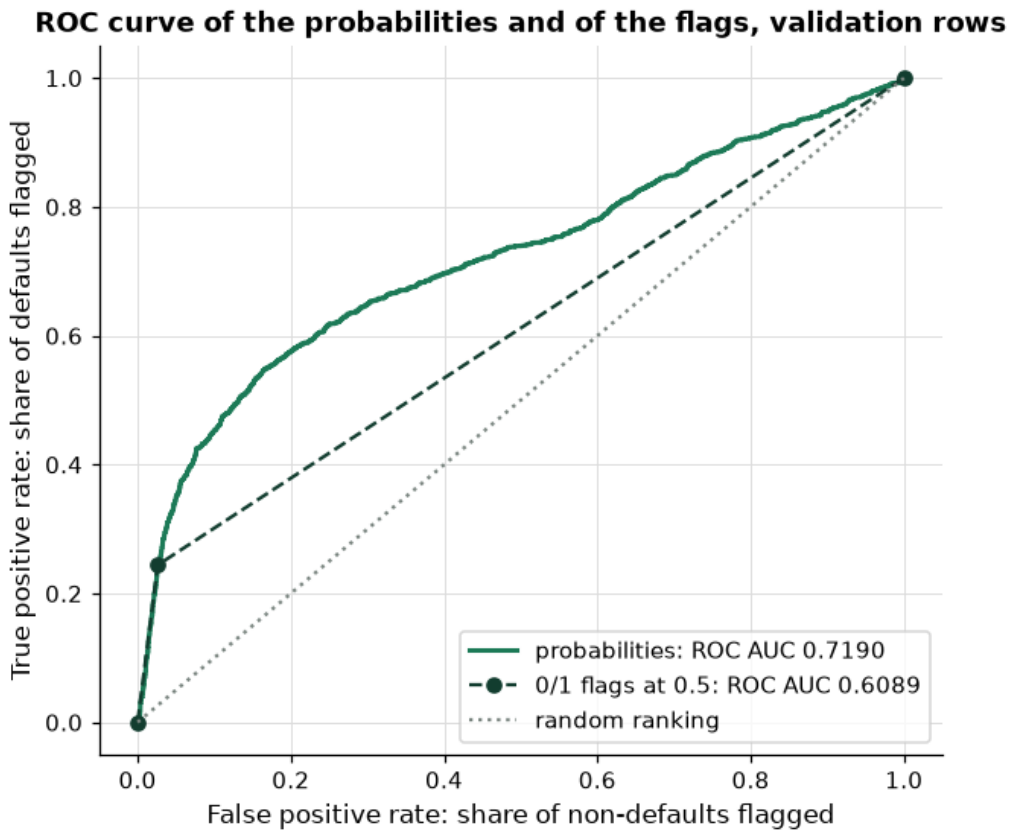
Threshold	Flagged	TPR	FPR
0.5	447	0.2442	0.0263
0.3	1,047	0.4476	0.0969
0.2	2,905	0.7046	0.4216
0.1	4,848	0.9020	0.7813

Up the curve is a lower threshold. The diagonal is a score with no information.

How AUC works

The number	The area under the ROC curve, from 0 to 1
As ranking	The chance that a random defaulter gets a higher probability than a random non-defaulter
The anchors	0.5 for the diagonal, a model that ranks at random; 1 if every default ranks above every non-default
No threshold	It uses only the order of the probabilities, so it compares models directly
What comes out	One number, unchanged by any rescaling that keeps the order

AUC as a ranking measure



0.7190

ROC AUC, probabilities

0.6089

the 0/1 flags at 0.5

One account that defaulted, one that did not: AUC is the chance the model ranks the defaulted one higher. Here, 71.90 percent of 6,201,071 pairs.

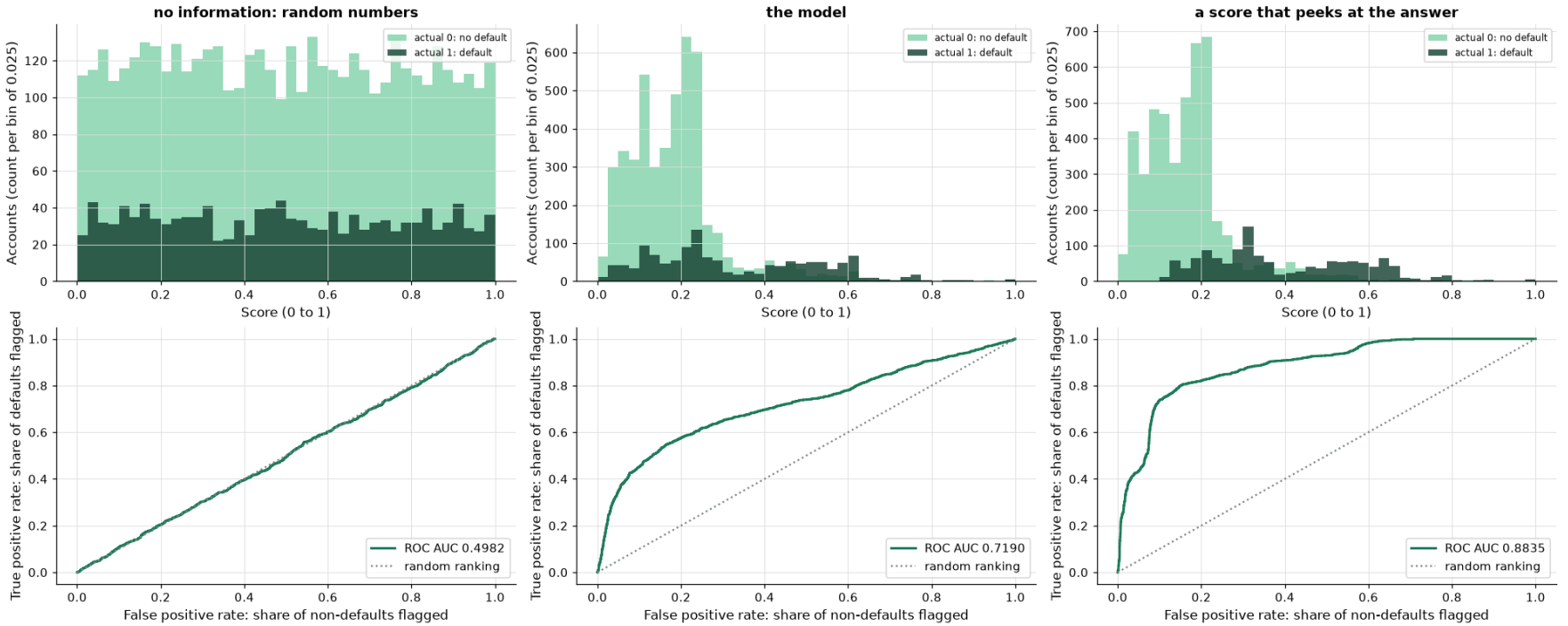
What AUC does not say

Not an accuracy, not a default rate

Nothing about any one threshold

Not whether a 0.3 means three in ten: squared, 0.7190

AUC as overlap

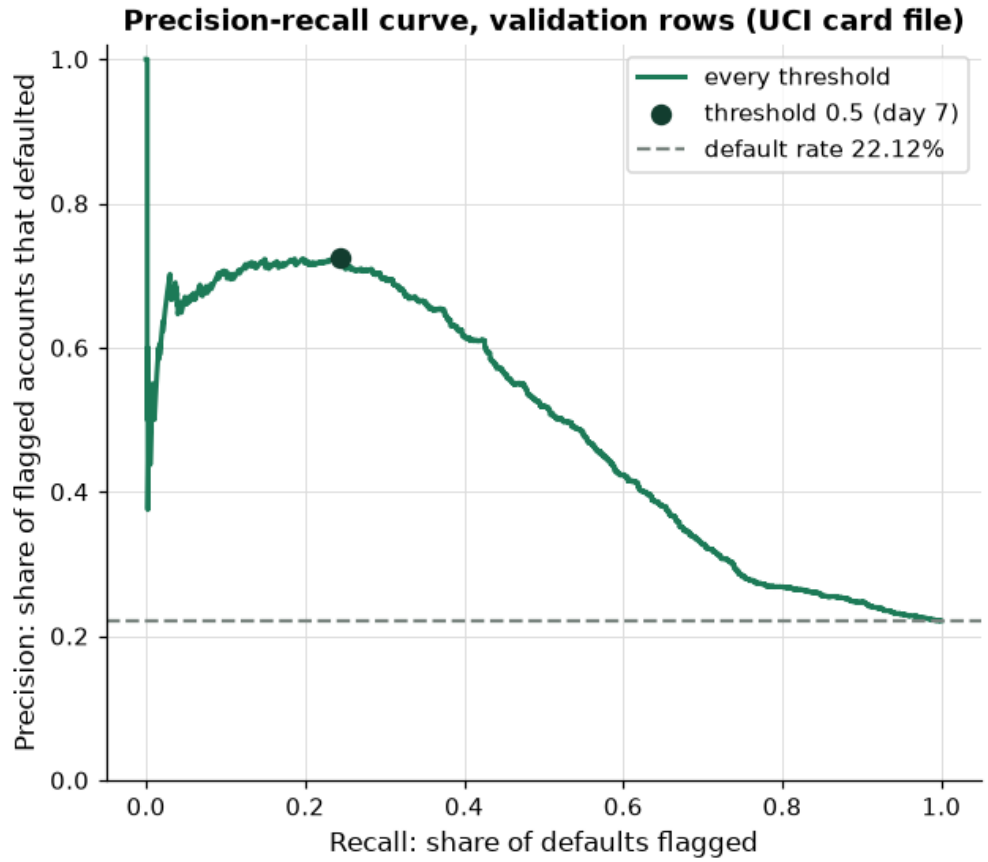


AUC 0.4982 for random numbers, 0.7190 for the model, 0.8835 for a score that has seen the answer: the less the groups overlap, the higher the curve.

How the precision-recall curve works

The gap in ROC	The false positive rate divides by all non-defaults, so a few hundred false flags barely move it
The desk asks	Of the accounts flagged, how many defaulted? That is precision
The curve	Precision against recall, one point per threshold: what each extra default costs in false flags
Average precision	Precision at each threshold, weighted by the recall that step adds
The floor	The default rate itself: flagging every account gives that precision

The precision-recall curve



0.4946

average precision

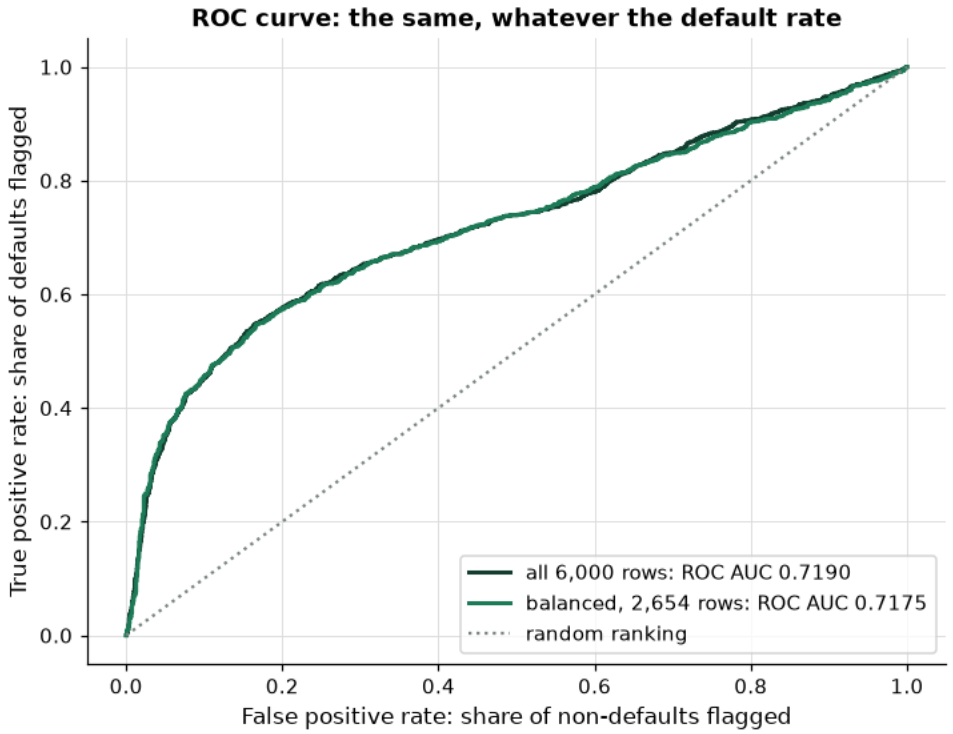
22.12%

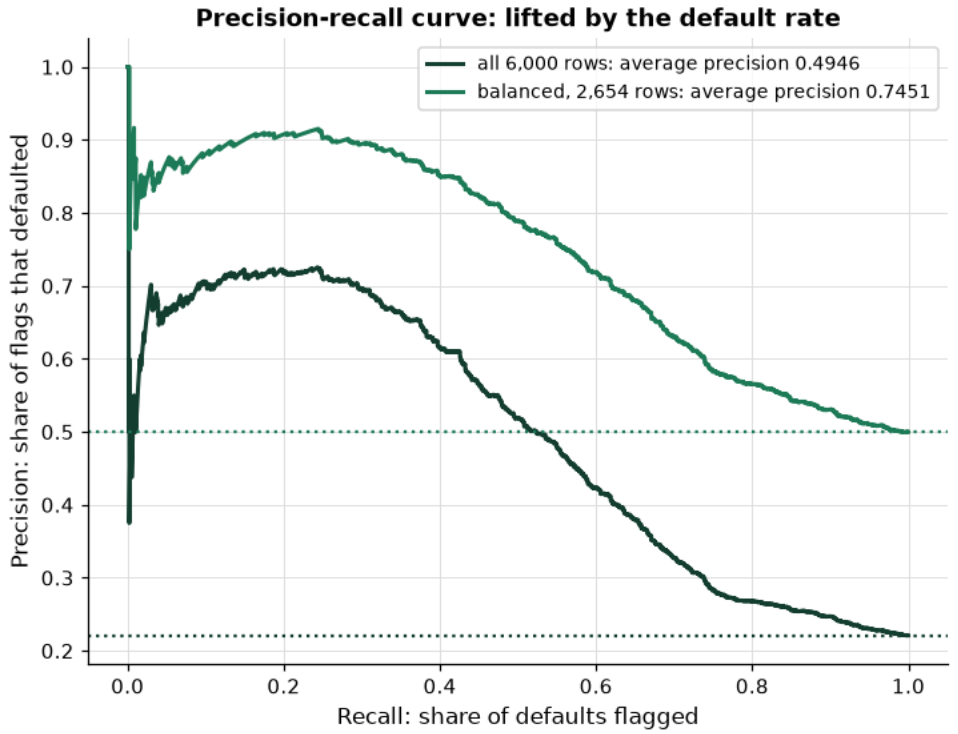
the floor: the default rate

Precision divides by the flagged accounts: of those flagged, how many defaulted? The false positive rate divides by every non-default.

Average precision is a step sum: each rise in recall times the precision there. Not a trapezoid.

Does the default rate change the curves?





ROC AUC 0.7190 and 0.7175; average precision 0.4946 and 0.7451. Quote average precision with the default rate beside it.

How a threshold is chosen

The trade	Every threshold trades defaults caught against accounts flagged that did not default
The rule first	Written down before looking, on rows that will not later report the result
The course's rule	Keep the thresholds whose validation recall reaches the target; take the highest
Why highest	A higher threshold flags fewer accounts, so the target is met with the fewest flags
What comes out	One threshold, applied unchanged to any other rows; a teaching convention, not a standard

Choosing a threshold on validation rows

The course's rule: mlkit.threshold_for_recall

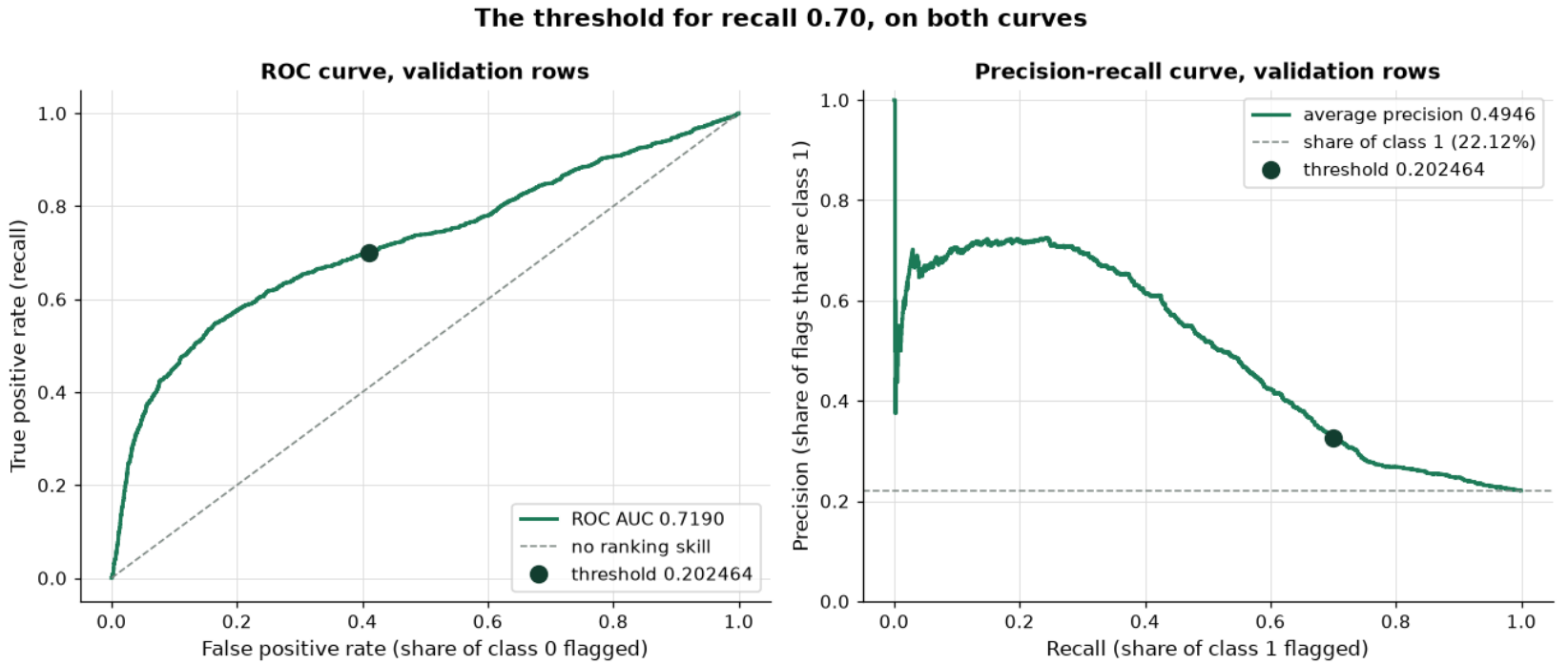
If the desk asked for recall of 70 percent: the highest threshold with validation recall at least 0.70.

Rule	Threshold	Flagged	Recall	Precision	F1
Day 7: 0.5	0.5000	447	0.2442	0.7248	0.3653
Round 0.2	0.2000	2,905	0.7046	0.3219	0.4419
Recall at least 0.70	0.2025	2,843	0.7001	0.3268	0.4456
Youden's J	0.2475	1,507	0.5494	0.4837	0.5145

Youden's J: where TPR minus FPR is largest (0.3829). Accuracy of flagging nothing: 0.7788; never used to choose.

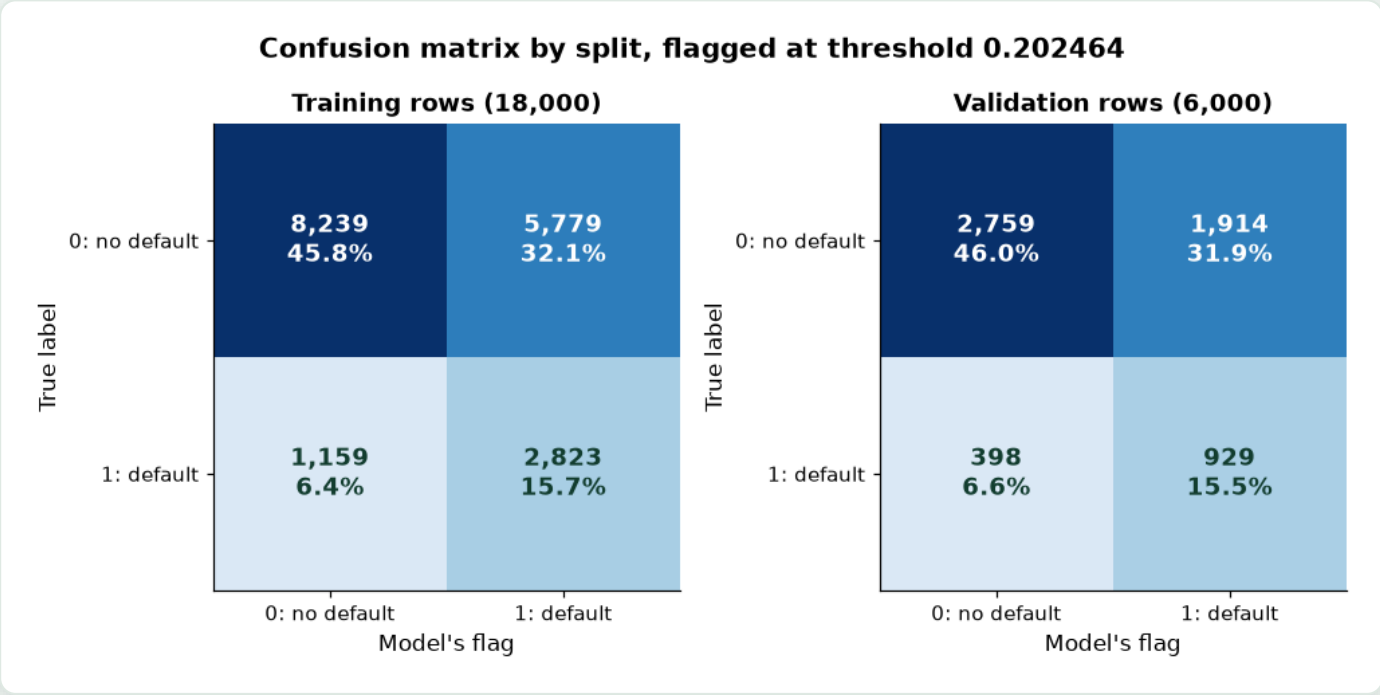
A teaching convention, not a lending standard. No threshold here is what a lender does or should do.

The chosen threshold on both curves



One threshold, one point on each curve: recall 0.7001, precision 0.3268. About one flagged account in three defaulted.

The chosen threshold, training and validation rows

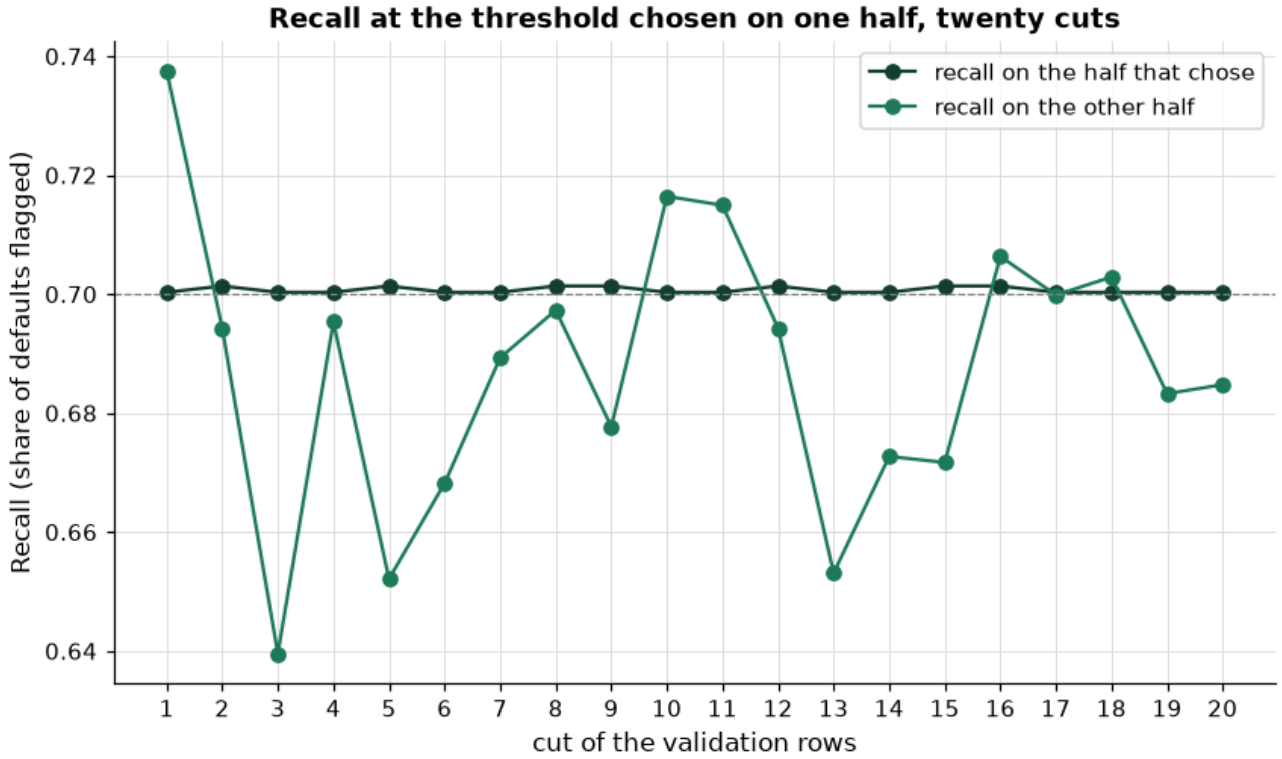


Rows	Recall	Precision	F1
training	0.7089	0.3282	0.4487
validation	0.7001	0.3268	0.4456

Read the panels as shares: the splits differ in size. The two rows are close.

The validation rows chose this threshold, so their recall is not the number to report.

Why the test rows are not used to choose



0.7003 to 0.7014

the half that chose

0.6395 to 0.7376

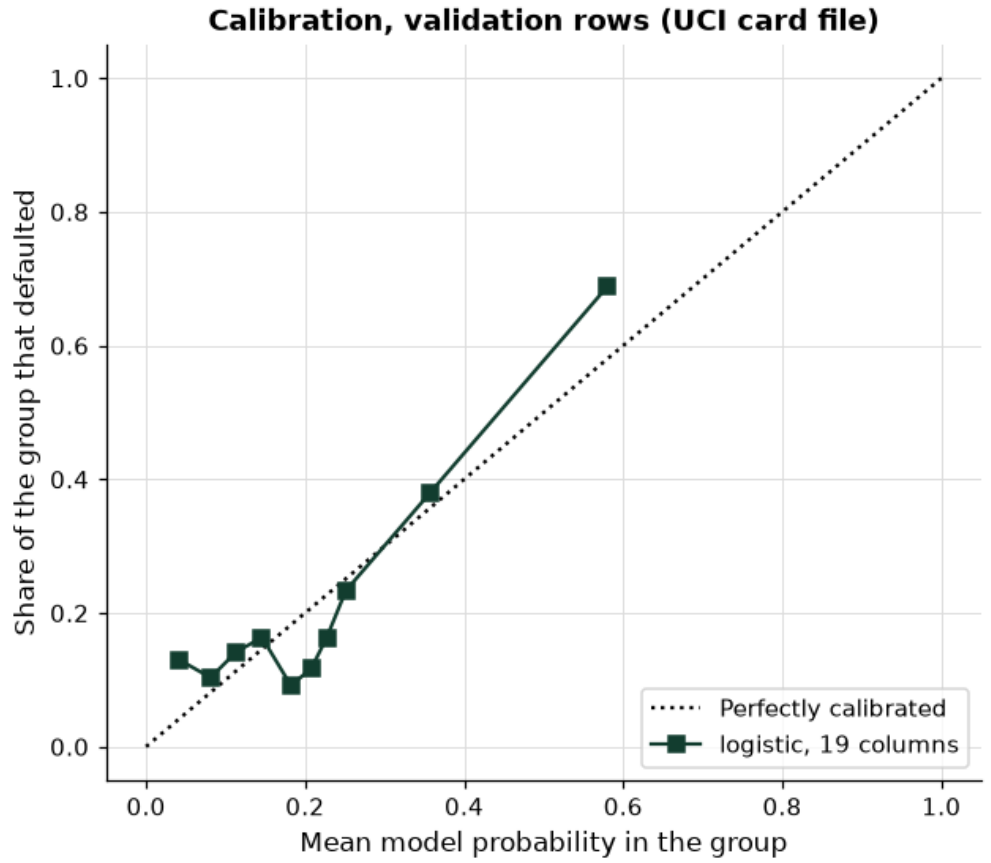
the other half: below 0.70 in 15 of 20

Chosen on the test rows, the reported recall would be 0.70 by construction. Day 10 opens them once.

How a calibration curve works

The question	Of the accounts given about 0.3, did about three in ten default?
Ten groups	Sort the accounts by probability into ten groups of equal size
The plot	Each group's mean probability against the share of its accounts that defaulted
The reading	On the diagonal the probabilities mean what they say; above it they understate, below it they overstate
You control	The number of groups, and which rows to use

Calibration: does a 0.3 mean three in ten?



Group	Mean probability	Defaulted
Lowest	0.0416	13.00%
Nearest 0.3	0.2512	23.33%
Top	0.5800	68.83%

Ten groups of about 600 accounts, sorted by probability. On the diagonal, a probability means what it says.

It ranks better than chance; its probabilities are not uniformly reliable as numbers. Recalibration: Course 5.

Excel side by side

Excel side by side (tested in Excel)

`=COUNTIFS(A2:A6001,1,B2:B6001,">="&D2)  (row 2)`	TP 324 at 0.5
`=F2/COUNTIF(A2:A6001,1)`	recall 0.2442
`=SORT(UNIQUE($B$2:$B$6001),,-1)  (M3)`	5,882 thresholds
`=COUNTIFS($A$2:$A$6001,1,$B$2:$B$6001,">="&M3#)/COUNTIF($A$2:$A$6001,1)  (O3)`	TPR, spilled
`=SUMPRODUCT((N3:N5884-N2:N5883),(O3:O5884+O2:O5883)/2)`	AUC 0.7190
`=MAXIFS(M3#,O3#,">=0.7")`	rule 0.2025
`=INDEX(M3#,MATCH(MAX(O3#-N3#),O3#-N3#,0))`	Youden 0.2475
`=SUMPRODUCT((O3:O5884-O2:O5883),P3:P5884)`	AP 0.4946

Excel has no ROC or AUC function: the probabilities come from Python; every step after them is a formula.

Save the day: the history of the choices

```
git log --oneline -- results.csv
  Experiment log: Youden's J threshold beside it, for comparison
  Experiment log: the threshold for recall 0.70, chosen on validation rows
  Experiment log: the logistic model at day 7's threshold 0.5
```

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
logistic_scaled at fixed 0.5,validation,0.7190,0.4946,0.5000,0.8123,0.7248,0.2442,0.3653
logistic_scaled at recall at least 0.70,validation,0.7190,0.4946,0.2025,0.6147,0.3268,0.7001,0.4456
logistic_scaled at Youden J,validation,0.7190,0.4946,0.2475,0.7707,0.4837,0.5494,0.5145
```

One row per choice, one commit per row. The two dashes keep only the commits that changed results.csv.

Why the threshold is what it is: the log shows the order of the choices and the commit behind each.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the threshold is chosen on the test rows, with a comment that says the model never saw them.

Code that runs is not the same as code that is right.

What you can do now

- The ROC curve: every threshold at once, each point a confusion matrix
- AUC 0.7190: pairs ranked the right way, nothing about one threshold
- Average precision 0.4946: a step sum over the precision-recall curve
- Recall at least 0.70 on validation rows: threshold 0.2025, a convention
- The rows that choose cannot report; calibration checks the numbers

Week 2, Day 4 Decision trees: a model made of questions, on the same validation rows